

Prompt Engineering

Fundamentals & Best Practices

Ek clean, non-repeated guide — AI Engineer ke daily kaam ke liye essential concepts, rules, aur debug workflow.

Prompt Anatomy

Zero / One / Few-shot

Structured Output

Debug Loop

Common Failures

1 Basics

Prompt ek "work order" ki tarah hota hai ek tez assistant ke liye. Achay prompts **Short, Specific, aur Testable** hote hain. Zaroorat se zyada politeness ya context avoid karein — **authoritative tone** rakhein. Goal: kam se kam interactions mein reliable output hasil karna.

2 Prompt ki Anatomy — 5 Core Components

1 Role

Model ka role define karein.

"You are a course tutor."

2 Task

Clear aur unambiguous kaam batayein.

"Summarize the five bullet notes into three actions."

3 Context

Sirf wahi info dein jo zaroori ho — irrelevant details avoid karein.

4 Rules & Constraints

Tone, style, length, include/avoid items specify karein.

5 Output Format

Response ka exact form batayein — JSON / Markdown / CSV.

Role: Course tutor

Task: Paanch bullet notes ko teen actions mein summarize karna

Rules: Spartan tone, 12 words se kam, no jargon

Output: Markdown bullets

Context: Team ki clarity, API delays, design mockups, customer feedback, documentation

3 Key Terminology

Term	Matlab
Token	Text ka unit (~4 characters)
Context Window	AI jitna text ek waqt mein process kar sakta hai — zyada context accuracy kam kar sakta hai
Temperature (API)	Low → focused & consistent · High → varied & creative
System message	Model ke liye persistent instructions
User message	Aap ka input
Assistant message	AI ka response
Zero-shot	Koi example nahi
One-shot	Ek example
Few-shot	2-5 examples

4 Five Core Rules

1 Task & Audience Specific Rahein

Task, audience, tone, style, prohibited terms clearly define karein.

Analogy: Language model ek GPS ki tarah hai — clear destination ke saath best kaam karta hai.

2 Context Ko Delimit Karein

Input ko clear, divided sections mein todein (dashes/headers). Isse read karna asaan hota hai, confusion kam hota hai.

3 Ek Strong Example Use Karein

Ek strong example, kai mediocre examples se behtar hai. Sirf variation/edge cases ke liye 2–3 examples use karein.

4 Structured Output

JSON → automation · Markdown → reports · CSV → tables. Sirf valid JSON, extra text nahi, unknown fields → null.

5 Pehle Plan, Phir Jawab

Multi-step/reasoning tasks mein pehle plan dikhwayein, phir final answer — verify karna asaan ho jata hai.

5 Quick Debug Loop

Prompt engineering **iterative** hai — best prompt usually kai iterations ke baad milta hai.

1 A/B Test Variations

Do versions compare karein; alag models/temperatures par test karein.

2 Isolate Variables

Ek waqt mein sirf ek cheez change karein.

Analogy: Audio mixing ke EQ knobs ki tarah — ek waqt mein ek hi knob ghumayein.

3 Golden Set Maintain Karein

Fixed test prompts rakhein taake performance compare ho sake aur regressions pakri ja sakein.

4 Performance Track Karein

Rubric ke against check karein: format, length, tone, accuracy — especially jab LLM kisi app/API ko power kar raha ho.

6 Common Failures & Fixes

Problem	Fix
Vague/ambiguous request	Role + audience + success criteria add karein
Messy output	Exact output format specify karein
Random tone	Style/tone constraint add karein
Output too long	Max token/length cap add karein
Too creative / off-topic	Temperature kam karein
Inconsistent answers	Structured prompt + examples use karein
Context ki kami	Relevant context provide karein
Conflicting instructions	Constraints ko align karein
Overloaded prompts	Bohat zyada opposing constraints na dein
Proper testing skip karna	Iteration/evaluation zaroori hai

7 Final Checklist

- ✓ Goal clearly define hai
- ✓ Role banaya gaya hai
- ✓ Audience clarify hai
- ✓ Sirf relevant context diya hai
- ✓ Ek strong example/structure diya hai
- ✓ Style/tone specify hai
- ✓ Scope define hai (kya ignore karna hai)
- ✓ Output format + length restrictions specify hain

ONE-LINE SUMMARY

Prompt engineering lambay prompts likhne ke baray mein **nahi** —
ye **clearer, testable, aur improvable** instructions likhne ke baray mein hai.
Weak prompts ek dafa nahi fail hotay — **silently, baar baar** fail hotay hain.